



# **HSC Mathematics Assessment Task 3 June 2011**

- Time allowed: 60 minutes plus 5 minutes reading time
- Answer each question on your own paper showing all necessary working.
- Start each question on a new page.
- Marks may not be awarded for untidy or poorly set out work.

| Topic                                          | Mark |
|------------------------------------------------|------|
| Trigonometric Functions                        | /25  |
| Applications of Calculus to the Physical World | /30  |

**TOTAL                    /55**

## TRIGONOMETRIC FUNCTIONS

### Question 1.

Marks

- a) Express  $\frac{2\pi}{3}$  radians in degrees. 1
- b) The angle subtended at the centre of a circle of radius 8 cm is 1.2 radians.  
Find:
- i) the length of the arc cut off by this angle 1
- ii) the area of the sector formed. 1
- c) Differentiate with respect to  $x$ :
- i)  $\sin 3x$  1
- ii)  $\tan \pi x$  1
- iii)  $x \cos x$  2
- iv)  $\sin^2 x$  2
- d) Find:
- i)  $\int \sin x \, dx$  1
- ii)  $\int \cos\left(2x + \frac{\pi}{3}\right) dx$  1
- iii)  $\int \sec 2x \tan 2x \, dx$  1
- iv)  $\int \frac{\sin x}{1 + \cos x} \, dx$  2

**Question 2. START A NEW PAGE**

- a) For the function  $y = 3 \sin 2x$ , find:
- i) its amplitude 1
  - ii) its period 1
- b) Solve  $2 \sin^2 x = \sin x$  for  $0 \leq x \leq 2\pi$  2
- c) i) On the same diagram, draw a neat sketch of  $y = \tan \pi x$  and  $y = 1 - x$  for  $0 \leq x \leq 2$  3
- ii) Hence determine the number of solutions to the equation  $\tan \pi x = 1 - x$  for  $0 \leq x \leq 2$ . 1
- d) The area under the curve  $y = \sec x$ , for  $0 \leq x \leq \frac{\pi}{4}$  is rotated about the  $x$  axis. What is the volume of the solid of revolution generated? 3

**APPLICATIONS OF CALCULUS TO THE PHYSICAL WORLD**

**Question 3. START A NEW PAGE**

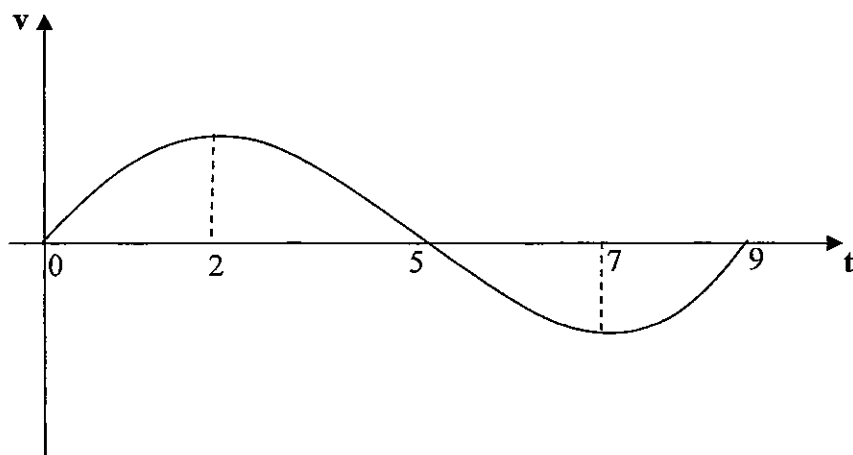
- a) A stone dropped into a still pond causes circular ripples on the surface. The area of disturbed water,  $t$  seconds after the stone hits the surface of the water is given by  $A = 4\pi t^2$
- i) Find an expression to represent the rate of change of the area of disturbed water. 1
  - ii) Find, in terms of  $\pi$ , the rate of change of the area of the disturbed water after 1 second. 1
- b) A particle moves on the  $x$  axis with velocity after  $t$  seconds given by  $v = 6 - 2t$  m/s. Initially it is at  $x=4$ .
- i) Find its position as a function of  $t$ . 2
  - ii) When is the particle at rest? 1
  - iii) Find the total distance travelled in the first 7 seconds. 2

- c) Water enters a container, initially empty, so that, after  $t$  minutes the volume  $V$  L of water in it is increasing at a rate of  $\frac{12t}{t^2 + 4}$  L/min.
- i) Show that  $V = 6 \ln\left(\frac{t^2 + 4}{4}\right)$  2
- ii) Find, to the nearest minute, the time taken for the container to hold 17 L. 2
- d) The population of a colony of insects at time  $t$  years is given by  $P = 40e^{\frac{1}{2}t}$ .
- i) Show that the growth is exponential. 1
- ii) What is the initial population? 1
- iii) What is the population after 10 years? 1
- iv) What is the rate of increase of the population after 10 years? 1

**Question 4. START A NEW PAGE**

- a) A particle P is moving along the  $x$  axis. Its position at time  $t$  seconds is given by  $x = 2 \sin t - t$ ,  $t \geq 0$ .
- i) Find an expression for the velocity of the particle. 1
- ii) In what direction is the particle moving at  $t = 0$ ? 1
- iii) Determine when the particle first comes to rest. 1
- iv) When is the acceleration negative for  $0 \leq t \leq 2\pi$ ? 2

b)



The above graph shows the velocity,  $v$  m/s, of a particle moving on a straight line, for  $0 \leq t \leq 9$ .

State all the times, or intervals of time, for which

- |      |                                                  |   |
|------|--------------------------------------------------|---|
| i)   | the particle is stationary                       | 1 |
| ii)  | the particle is moving in the positive direction | 1 |
| iii) | the acceleration is positive                     | 1 |
| iv)  | the particle is slowing down.                    | 1 |

- c) The mass,  $M$  g, of a radioactive element present in a substance after  $t$  years is given by  $M = M_0 e^{-kt}$ , where  $M_0$  is the initial mass and  $k$  is a constant. The half-life period of the element is 100 years.

- |      |                                                                     |   |
|------|---------------------------------------------------------------------|---|
| i)   | Show that $k = \frac{\ln 2}{100}$                                   | 2 |
| ii)  | How long will it take for 9 g to reduce to 2 g?                     | 2 |
| iii) | What percentage of an original mass will be present after 32 years? | 2 |

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE:

$$\ln x = \log_e x, x > 0$$

Question 1

$$a) \quad \frac{2\pi}{3} = \frac{2 \times 180}{3} = 120$$

$$b) \quad i) \quad l = r\theta$$

$$l = 8 \times 1.2$$

$$= 9.6 \text{ cm}$$

$$ii) \quad A = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 8^2 \times 1.2$$

$$= 38.4 \text{ cm}^2$$

$$iv) \quad \int \frac{\sin u \, du}{1 + \cos u}$$

$$= -\ln(1 + \cos u) + c$$

$$c) \quad i) \quad y = \sin 3x$$

$$y' = 3 \cos 3x$$

$$ii) \quad y = \tan \pi x$$

$$y' = \pi \sec^2 \pi x$$

$$iii) \quad y = x \cos x$$

$$y' = -x \cdot \sin x + \cos x$$

$$iv) \quad y = \sin^2 x$$

$$y' = 2 \sin x \cos x$$

$$d) \quad i) \quad \int \sin x \, dx = -\cos x + c$$

$$ii) \quad \int \cos \left( 2x + \frac{\pi}{3} \right) dx = \frac{1}{2} \sin \left( 2x + \frac{\pi}{3} \right) + c$$

$$iii) \quad \int \sec 2x \tan 2x \, dx = \frac{1}{2} \sec 2x + c$$

2  
a)  $y = 3 \sin 2x$

i)  $a = 3$

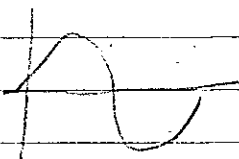
ii)  $T = \frac{2\pi}{2} = \pi$

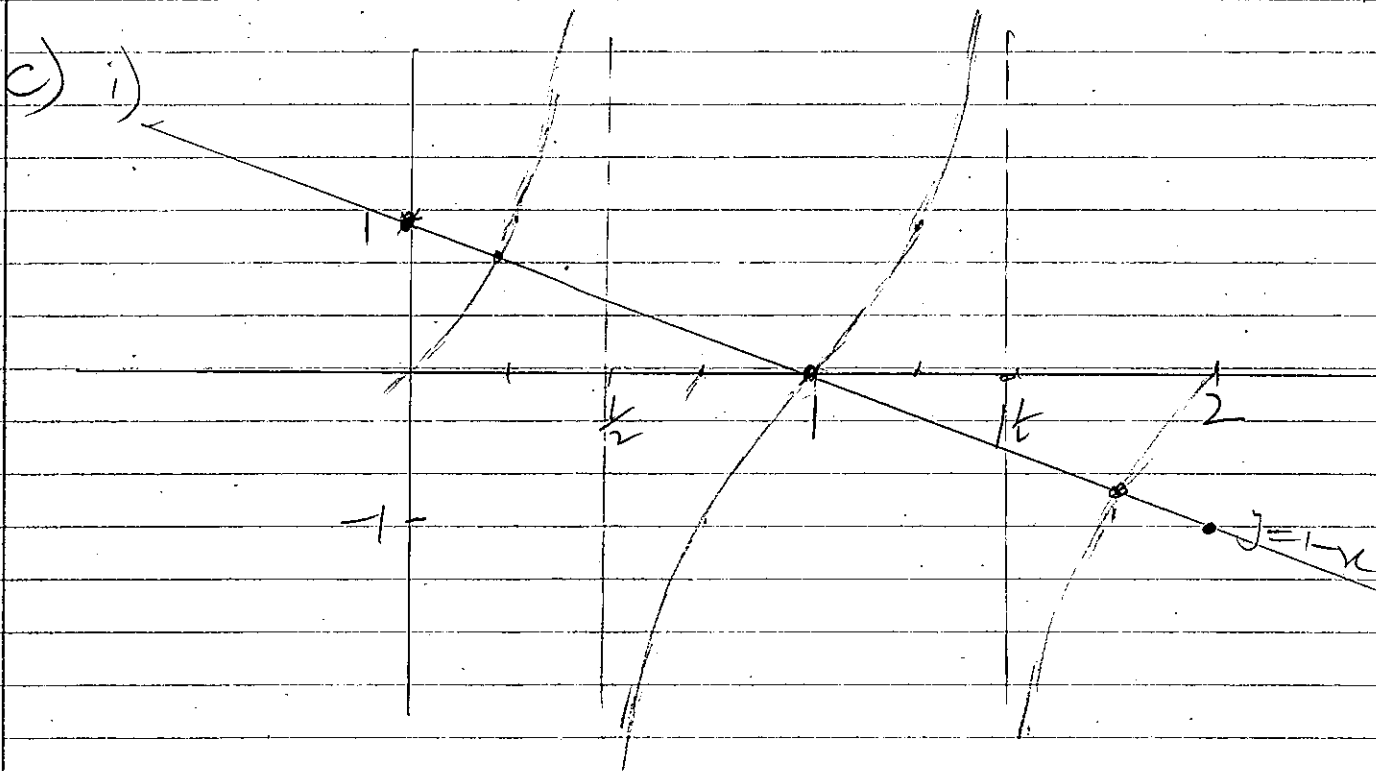
b)  $2 \sin^2 x = \sin x$

$$2 \sin^2 x - \sin x = 0$$

$$\sin x (2 \sin x - 1) = 0$$

$$\sin x = 0 \quad \text{or} \quad \sin x = \frac{1}{2}$$

  $x = 0, \pi, 2\pi$        $x = \frac{\pi}{6}, \frac{5\pi}{6}$



ii) there are 3 solutions.

d) 
$$V = \pi \int_0^{\pi/4} \sec^2 x \, dx = \pi \left[ \tan x \right]_0^{\pi/4}$$

$$= \pi \times 1 - \pi \times 0 = \pi u^3$$

### Question 3

a)  $A = 4\pi t^2$

i)  $\frac{dA}{dt} = \underline{\underline{8\pi t}}$

ii)  $t=1$ ,  $\frac{dA}{dt} = \underline{\underline{8\pi}} \text{ m}^2/\text{s}$

b)  $v = 6 - 2t$

i)  $x = 6t - t^2 + c$

$t=0$ ,  $x=4$

$4 = 0 - 0 + c$

$\therefore x = 6t - t^2 + 4$

ii)  $v=0$  for at rest

$0 = 6 - 2t$

$\underline{\underline{t=3}}$

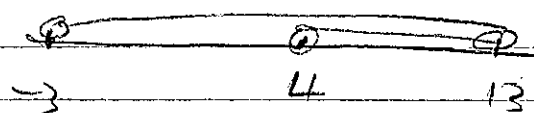
At 3 secs particle is at rest.

iii) Dist  $t=0$   $x=4$

$t=7$   $x=-3$

When  $t=3$ ,  $x=13$

Distance travelled:



$13 - 4 = 9$

$d = 2 \times 9 + 7$

$= 25 \text{ metres}$

2c)

$$\frac{dV}{dt} = \frac{12t}{t^2+4}$$

$$V = \int \frac{12t}{t^2+4} dt$$

$$V = 6 \ln(t^2+4) + C$$

Initially  $t=0$ ,  $V=0$ .

$$0 = 6 \ln 4 + C$$

$$C = -6 \ln 4$$

$$V = 6 \ln(t^2+4) - 6 \ln 4$$

$$V = 6 \ln\left(\frac{t^2+4}{4}\right) \quad \text{as required.}$$

ii)

$$17 = 6 \ln\left(\frac{t^2+4}{4}\right)$$

$$\frac{17}{6} = \ln \frac{t^2+4}{4}$$

$$t^2+4 = 4e^{\frac{17}{6}} \Rightarrow t^2 = 4e^{\frac{17}{6}} - 4$$

$$t = \pm \sqrt{64.008} \approx 8 \text{ mins}$$

d i)

$$P = 40e^{\frac{t}{2}}$$

$$\frac{dP}{dt} = 40 \times \frac{1}{2} e^{\frac{t}{2}}$$

$$= 20e^{\frac{t}{2}}$$

$$= \frac{1}{2} P$$

rate proportional to  $P$

ii)

$$t=0 \quad P = 40e^0 = 40$$

iii)

$$t = 10$$

$$P = 40 e^{\frac{1}{2} \times 10}$$
$$= 5936.52$$

$$P \approx 5937$$

iv)

$$\frac{dP}{dt} = 40 \times \frac{1}{2} e^{\frac{1}{2}t}$$

$$= \frac{1}{2} (40 \times e^5)$$

$$= 2968 \text{ insects / year}$$

### Question 4

a)  $x = 2\sin t - t$

i)  $v = 2\cos t - 1$

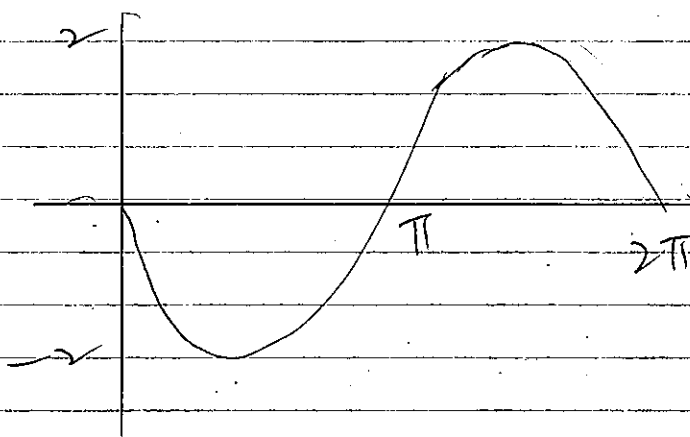
ii)  $t=0, \quad v = 2 - 1$   
 $= 1$

Moving to the right

iii)  $2\cos t - 1 = 0$   
 $\cos t = \frac{1}{2}$

$t = \frac{\pi}{3}, \quad \dots$

iv)  $\ddot{x} = -2\sin t$



Acc is negative for

$$0 < t < \pi$$

b) i) Stationary  $t=0, 5, 9$

ii)  $0 < t < 5$

iii) Gradient of  $v$  graph is positive  
 $0 < t < 2$  and  $7 < t < 9$

iv)  $v$  &  $\ddot{x}$  in opposite directions  
 $2 < t < 5$  and  $7 < t < 9$   
 $v > 0, \ddot{x} < 0$        $v < 0, \ddot{x} > 0$

c)  $M = M_0 e^{-kt}$

$$t = 100, \quad M = \frac{M_0}{2}$$

$$\frac{M_0}{2} = M_0 e^{-100k}$$

$$0.5 = e^{-100k}$$

$$2 = e^{100k}$$

$$100k = \ln 2$$

$$k = \frac{\ln 2}{100}$$

ii)  $M = M_0 e^{-\frac{\ln 2}{100} t}$

$$2 = 9 e^{-\frac{\ln 2}{100} t}$$

$$\ln \frac{2}{9} = -\frac{\ln 2}{100} t$$

$$t = \ln \frac{9}{2} \div \left( -\frac{\ln 2}{100} \right)$$

$$t = \frac{\ln \frac{2}{9}}{-\frac{\ln 2}{100}}$$

$$= 216.99$$

$$t = 217 \text{ years}$$

iii)

$$M = M_0 e^{-\frac{\ln 2}{100} \times 32}$$

$$M = 0.801 M_0$$

80.1% of original mass is present after 32 years.

80.1